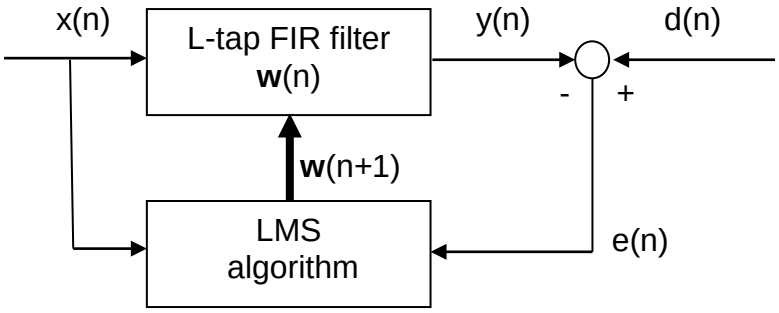
Standard LMS algorithm



Parameters L = filter length
μ = step size

Init

$$\mathbf{x}(0) = \begin{bmatrix} x(0) \\ \vdots \\ x(-(L-1)) \end{bmatrix} = \mathbf{0}$$

$$\mathbf{w}(0) = \begin{bmatrix} w_0(0) \\ \vdots \\ w_{L-1}(0) \end{bmatrix} = \mathbf{0}$$

$n = 1$

Loop **Get data**

$$\mathbf{x}(n) = \begin{bmatrix} x(n) \\ \vdots \\ x(n-(L-1)) \end{bmatrix}$$

$d(n)$

Predict $\hat{d}(n) = y(n) = \mathbf{w}^T(n-1)\,\mathbf{x}(n)$

Error $e(n) = d(n) - \hat{d}(n)$

Update filter $\mathbf{w}(n) = \mathbf{w}(n-1) + \mu\, e(n)\, \mathbf{x}(n)$

Iterate $n = n + 1$

End

Computational complexity $\mathcal{O}(L)$

Performance of standard LMS algorithm

Convergence / stability constraint

Result

Under stationary conditions the standard LMS algorithm will converge towards the minimum of the cost function $\xi(n) = E[e^2(n)]$ if the step size μ is constrained by

$$0 < \mu < \frac{2}{\lambda_{max}}$$

where λ_{max} is the largest eigenvalue of the symmetric and positive semi-definite autocorrelation matrix $\mathbf{R}_{xx}(n) = E[\mathbf{x}(n)\mathbf{x}^T(n)]$ (having only real, non-negative eigenvalues)

Practical version of result

Since $\lambda_{max} \le \sum_{i=1}^{L} \lambda_i = \text{trace}(\mathbf{R}_{xx}) = L\, r_{xx}(0) = L\, P_x$, where P_x is the power of $x(n)$, the convergence constraint can be reformulated in more practical terms as

$$0 < \mu < \frac{2}{L\, \hat{P}_x(n)}$$

where $\hat{P}_x(n) = \frac{1}{K}\sum_{m=n-(K-1)}^{n} x^2(m)$ is an estimate of signal power

A reasonable first attempt is setting $\mu = \frac{1}{10} \cdot \frac{2}{L\, \hat{P}_x(n)}$

Argument (somewhat technical)

$$\begin{aligned}\mathbf{w}(n+1) &= \mathbf{w}(n) - \tfrac{\mu}{2}\,\nabla\xi(n) = \mathbf{w}(n) - \tfrac{\mu}{2}\,[2\mathbf{R}_{xx}(n)\mathbf{w}(n) - 2\mathbf{r}_{dx}(n)] \\ &= \mathbf{w}(n)\,[I - \mu\mathbf{R}_{xx}(n)] + \mu\mathbf{r}_{dx}(n)\end{aligned}$$

Subtracting on both sides the optimal Wiener Hopf solution $\mathbf{w}_{opt}(n) = \mathbf{R}_{xx}^{-1}(n)\,\mathbf{r}_{dx}(n)$
and defining $\Delta\mathbf{w}(n) = \mathbf{w}(n) - \mathbf{w}_{opt}(n)$ gives

$$\Delta\mathbf{w}(n+1) = [I - \mu\mathbf{R}_{xx}(n)]\,\Delta\mathbf{w}(n)$$

Next, since $\mathbf{R}_{xx}(n)$ is symmetric and positive semi-definite, its eigenvalue decomposition can be written as

$$\mathbf{R}_{xx}(n) = V\,\Lambda\,V^T \quad \text{where} \quad \Lambda = \begin{bmatrix}\lambda_1 & 0 & 0\\ 0 & \ddots & 0\\ 0 & 0 & \lambda_L\end{bmatrix},\ \lambda_i \ge 0 \quad \text{and} \quad VV^T = I,\ \ |V| = 1$$

Thus

$$\begin{aligned}\Delta\mathbf{w}(n+1) &= [I - \mu\, V\,\Lambda\,V^T]\,\Delta\mathbf{w}(n) = V[I - \mu\,\Lambda\,]V^T\Delta\mathbf{w}(n) \\ &= V\begin{bmatrix}1-\mu\lambda_1 & 0 & 0\\ 0 & \ddots & 0\\ 0 & 0 & 1-\mu\lambda_L\end{bmatrix}V^T\Delta\mathbf{w}(n)\end{aligned}$$

Finally, it can be seen that the condition for decreasing $\Delta\mathbf{w}(n)$, i.e. $\Delta\mathbf{w}(n) \to \mathbf{0}$ for $n \to \infty$, is

$$|1 - \mu\lambda_i| < 1 \quad \text{or} \quad 0 < \mu < \frac{2}{\lambda_i} \quad \text{for} \quad i = 1, \dots, L \quad \text{implying} \quad 0 < \mu < \frac{2}{\lambda_{max}}$$

Convergence speed

Obviously, the speed of convergence is increased by increasing the step size μ, which, however, decreases the margin of algorithm stability.

From above the convergence toward optimal filter coefficients is exponential

$$\Delta w_i(n) = c(1-\mu\lambda_i)^n \qquad \text{for} \quad i = 1, \dots, L$$

Thus for each eigenvalue (”mode”) a time constant of convergence, τ_i, can be defined as, e.g.

$$(1-\mu\lambda_i)^{\tau_i} = \frac{1}{e}$$

or

$$\tau_i = -\frac{1}{\log(1-\mu\lambda_i)} \cong \frac{1}{\mu\lambda_i} \qquad \text{(if, for stability reasons, } \mu \ll \frac{1}{\lambda_i}\text{)}$$

The dominant (slowest) time constant for overall convergence will therefore be

$$\tau = \max\{\tau_i\} = \frac{1}{\mu\lambda_{min}}$$

Finally, since the step size is constrained by

$$0 < \mu < \frac{2}{\lambda_{max}} \qquad \text{or} \qquad \mu = \alpha \cdot \frac{2}{\lambda_{max}} \quad \text{with} \quad 0 < \alpha < 1,$$

the dominant time constant can be written as

$$\tau = \frac{1}{2\alpha}\frac{\lambda_{max}}{\lambda_{min}}$$

This shows that, for a fixed stability margin (value of α), the speed of convergence is determined by the spread of the eigenvalues and more precisely on the ratio between the biggest and the smallest eigenvalue (also known as the “condition number” for $\mathbf{R}_{xx}$).

Stationary and excess mean square error

Under stationary conditions, after convergence of the algorithm, the filter coefficients are characterized by

$$\Delta\mathbf{w}(n) \cong \mathbf{0} \qquad \text{or} \qquad \mathbf{w}(n) \cong \mathbf{w}_{opt} = \mathbf{R}_{xx}^{-1}\,\mathbf{r}_{dx}$$

However, since the algorithm is still iterating – to adapt to potential changes in signal and system characteristics over time – there will still be fluctuations in $\mathbf{w}(n)$

The mean square error, $\xi = E[e^2(n)]$, will therefore consist of two parts; one part, ξ_{min}, from the minimum MSE due to the Wiener Hopf solution, $\mathbf{w}_{opt}$, and one part, ξ_{excess}, coming from filter coefficient fluctuations away from $\mathbf{w}_{opt}$
Thus

$$\xi_{stationary} = E[e^2(n)] = \xi_{min} + \xi_{excess}$$

The first part of the MSE is

$$\xi_{min} = E[d^2(n)] - \mathbf{w}_{opt}^{T}\mathbf{r}_{dx} = E[d^2(n)] - \mathbf{r}_{dx}^{T}\mathbf{R}_{xx}^{-1}\mathbf{r}_{dx}$$

How close this part is to zero depends on the statistical properties of the signals $x(n)$ and $d(n)$ and on the filter length L. It should *not*, however, depend on the step size μ or the signal power P_x.

The extra part of the MSE due to filter coefficient fluctuations can be found to be

$$\xi_{excess} \cong \frac{1}{2}\,\mu\,L\,P_x\,\xi_{min}$$

However, since the step size for stability reasons is constrained by $\mu < \frac{2}{L\,P_x(n)}$

$$\xi_{excess} < \frac{1}{2}\,\frac{2}{L\,P_x(n)}\,L\,P_x\,\xi_{min} = \xi_{min}$$

Thus, the dominant part of $\xi_{stationary}$ will be ξ_{min}.

The ratio between the two parts of $\xi_{stationary}$ is called the "misadjustment", $\mathcal{M}$

$$\mathcal{M} \overset{\text{def}}{=} \frac{\xi_{excess}}{\xi_{min}} \cong \frac{1}{2}\,\mu\,L\,P_x$$

If $0 < \mu \ll \frac{2}{L\,P_x(n)}$ the misadjustment will be $0 < \mathcal{M} \ll 1$